# Synoptic Question 31: The Physics of a Washing Machine Spin Cycle

## Reading Passage

*Read the following article and then answer the questions that follow.*

An automatic washing machine uses mechanical agitation and thermal energy to clean clothes. A key part of the process is the spin cycle, designed to remove excess water through rapid rotation. The inner drum, containing the wet clothes, is accelerated to a high angular velocity. Water is forced out through perforations in the drum by the large centripetal force required to keep the water moving in a circle.

The entire drum assembly (including the motor and outer tub) is typically suspended within the machine's casing by a system of springs and dampers (shock absorbers), as shown schematically in Fig. 31.1. This suspension allows the assembly to vibrate during the spin cycle, particularly if the load of clothes is unevenly distributed (unbalanced).

Every mechanical system has one or more natural frequencies at which it tends to vibrate with large amplitude if driven at that frequency. This phenomenon is called **resonance**. If the rotational frequency of the drum during the spin cycle matches a natural frequency of the suspended assembly, violent vibrations can occur, potentially damaging the machine or causing excessive noise. Washing machine designs incorporate suspension systems with specific stiffness and damping characteristics to mitigate these vibrations and avoid resonance at the main operating speeds. Often, the machine accelerates rapidly through lower speeds that might correspond to natural frequencies. Damping, provided by the shock absorbers, dissipates vibrational energy as heat.

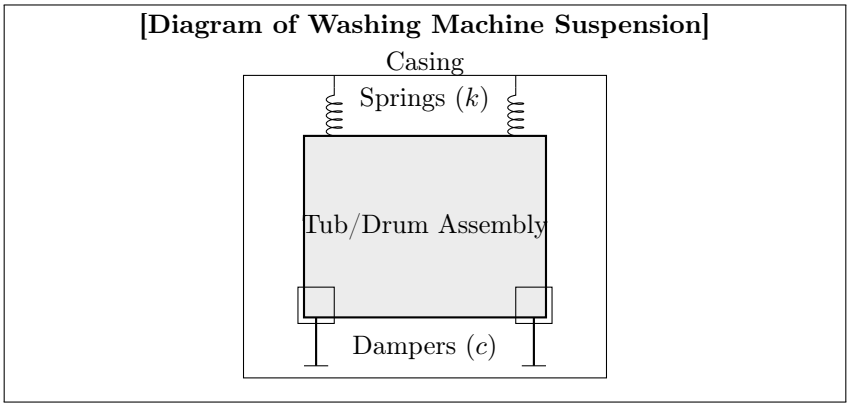


Figure 1: Simplified schematic of drum assembly suspension.

## Data:

Data for a typical front-loading washing machine during its spin cycle:

Table 1: Data for Washing Machine Spin Cycle Analysis

| **Parameter** | **Value** |
|---|---|
| Inner radius of drum, $r$ | 0.25 m |
| Total mass of suspended assembly (tub, drum, motor, wet clothes), $m_{tub}$ | 65.0 kg |
| Total effective spring constant of suspension, $k_{susp}$ | $2.80 \times 10^4$ N m$^{-1}$ |
| Maximum spin speed, $\omega_{max,rpm}$ | 1400 rpm |
| Time taken to accelerate from rest to max spin speed, $t_{accel}$ | 8.0 s |
| Moment of inertia of loaded drum (approximated), $I_{drum}$ | 1.80 kg m$^2$ |
| Resistive torque during spin (friction, damping), $\tau_{resist}$ | 2.50 Nm |
| Efficiency of motor, $\eta_{motor}$ | 70% |
| Mass of unbalanced load clump, $m_{unb}$ | 0.50 kg |

## Questions

(a) Define **natural frequency** and **resonance**.

(b) The suspended tub assembly can oscillate vertically on the springs.

  (i) Calculate the natural frequency $f_n$ (in Hz) of vertical oscillation for the loaded tub assembly.

  (ii) Convert the maximum spin speed (1400 rpm) into a rotational frequency $f_{spin}$ in Hz.

  (iii) Compare $f_n$ and $f_{spin}$. Explain why it is crucial for a washing machine design that these two frequencies are significantly different.

  (iv) The machine accelerates through various speeds before reaching $f_{spin}$. Suggest why the machine might be programmed to accelerate particularly quickly through the speed range around $f_n$.

(c) Consider the drum rotating at its maximum spin speed.

  (i) Calculate the angular velocity $\omega_{max}$ in rad $s^{-1}$.

  (ii) Calculate the magnitude of the centripetal acceleration experienced by clothes at the drum's inner radius. Express your answer in terms of $g$.

(d) During acceleration to maximum spin speed:

  (i) Assuming constant angular acceleration $\alpha$, calculate its value.

  (ii) Calculate the net torque $\tau_{net}$ required to produce this acceleration.

  (iii) Calculate the total torque $\tau_{motor}$ that the motor must provide during acceleration, considering the resistive torque.

(e) Consider the energy required during the spin cycle.

  (i) Calculate the rotational kinetic energy stored in the drum at maximum spin speed.

  (ii) Calculate the mechanical power $P_{mech}$ the motor must provide to maintain the maximum spin speed against the resistive torque.

  (iii) Calculate the electrical power $P_{elec}$ consumed by the motor to maintain this speed.

(f) An unbalanced load occurs when clothes gather unevenly, modelled here as a 0.50 kg mass clump stuck to the inside radius of the drum.

  (i) Calculate the magnitude of the unbalanced force generated by this mass rotating at the maximum spin speed.

  (ii) Describe qualitatively how the suspension system (springs and dampers) responds to this rotating unbalanced force. Explain the role of the dampers.

## Solutions

(a) **Natural Frequency ($f_n$):** The frequency at which a system oscillates when disturbed from its equilibrium position and allowed to oscillate freely (without any driving or damping forces).
**Resonance:** The phenomenon that occurs when a system is driven by a periodic external force at a frequency equal or very close to its own natural frequency, resulting in a large increase in the amplitude of oscillation.

(b) (i) For a mass-spring system, $f_n = \frac{1}{2\pi}\sqrt{\frac{k_{susp}}{m_{tub}}}$.
$f_n = \frac{1}{2\pi}\sqrt{\frac{2.80\times10^4}{65.0}} = \frac{1}{2\pi}\sqrt{430.8} = 3.30$ Hz.

(ii) $f_{spin} = 1400 \text{ rpm} \times \frac{1 \text{ min}}{60 \text{ s}} = 23.33 \text{ Hz} \approx 23.3 \text{ Hz}$.

(iii) $f_n(3.30 \text{ Hz}) \ll f_{spin}(23.3 \text{ Hz})$. It is crucial that the main operating frequency ($f_{spin}$) is far from the natural frequency ($f_n$) to avoid resonance, which would cause large amplitude, noisy, and potentially damaging vibrations.

(iv) Accelerating quickly through the speed range near $f_n$ (approx 3.3 Hz $\times$ 60 $\approx$ 200 rpm) minimizes the time spent driving the system at its resonant frequency. This prevents the amplitude of vibrations from building up significantly during start-up or slow-down.

(c) (i) $\omega_{max} = f_{spin} \times 2\pi = 23.33 \times 2\pi = 146.6 \text{ rad s}^{-1} \approx 147 \text{ rad s}^{-1}$.

(ii) $a_c = r\omega_{max}^2 = (0.25)(146.6)^2 = 5373 \text{ m s}^{-2}$.
In terms of $g$: $\frac{a_c}{g} = \frac{5373}{9.81} = 548$. So, $a_c \approx 550g$.

(d) (i) $\alpha = \frac{\omega_f - \omega_i}{t_{accel}} = \frac{146.6-0}{8.0} = 18.33 \text{ rad s}^{-2} \approx 18.3 \text{ rad s}^{-2}$.

(ii) $\tau_{net} = I_{drum}\alpha = (1.80)(18.33) = 33.0$ Nm.

(iii) The motor must provide torque to accelerate the drum ($\tau_{net}$) AND overcome friction ($\tau_{resist}$).
$\tau_{motor} = \tau_{net} + \tau_{resist} = 33.0 + 2.50 = 35.5$ Nm.

(e) (i) $KE_{rot} = \frac{1}{2}I_{drum}\omega_{max}^2 = \frac{1}{2}(1.80)(146.6)^2 = 19343 \text{ J} \approx 19.3 \text{ kJ}$.

(ii) To maintain constant speed, the motor torque must equal the resistive torque.
$P_{mech} = \tau_{resist}\omega_{max} = (2.50)(146.6) = 366.5 \text{ W} \approx 367 \text{ W}$.

(iii) $P_{elec} = \frac{P_{mech}}{\eta_{motor}} = \frac{366.5}{0.70} = 523.6 \text{ W} \approx 524 \text{ W}$.

(f) (i) The unbalanced mass $m_{unb}$ moves in a circle of radius $r$. It requires a centripetal force $F_{unb} = m_{unb}r\omega_{max}^2$.
$F_{unb} = (0.50)(0.25)(146.6)^2 = 2687 \text{ N} \approx 2690 \text{ N}$.
This rotating force is transmitted to the tub assembly.

(ii) The rotating unbalanced force acts as a periodic driving force on the suspended tub assembly, causing it to vibrate. The springs provide the restoring force for this vibration. The dampers (shock absorbers) exert a force that opposes the velocity of the vibration, dissipating the vibrational energy as heat. This limits the amplitude of the vibrations, especially if the driving frequency were close to the natural frequency, preventing excessive shaking of the machine.